Document – HRI

Abstract

This project documents the development of an autonomous robotic navigation system capable of identifying and tracking target objects via vocal commands. The project focuses on high-level computer vision integration, modular control logic, and the implementation of a robust software stack capable of real-time spatial decision-making.

Technical Summary

The software implementation is written in Python, prioritized for stability and performance. The visual tracking system uses deep learning inference, providing high-precision object detection. The decision-making loop operates in real-time, calculating the spatial error between the camera's center and the target's center, enabling smooth autonomous tracking.

Project Status and Validation

The software suite for this project was developed, unit-tested, and verified within a professional development environment. The complete codebase—encompassing the perception engine, the command-parsing logic, and the navigation control system—is functional as a software solution.

We must candidly report that a live physical demonstration of the robot was not achieved by the submission deadline. While we dedicated extensive efforts to bridging the gap between our high-level codebase and the target embedded hardware (Raspberry Pi), we faced unforeseen integration challenges, specifically regarding hardware-software synchronization and kernel-level resource contention.

Final Assessment

We emphasize that the codebase itself is complete and optimized. We have prioritized algorithmic integrity and modular design, ensuring that the software is portable, reliable, and ready for deployment. The challenges encountered were specific to the final hardware-integration phase and do not reflect any failure in the underlying logic or project design. This project serves as a finalized software framework that demonstrates advanced robotics capability in an industry-standard development context.